

Quasi experimental (Pre/post): Leadership and management program for Nurses leaders

PI: Amira Yahya AL Nasri

The aims of the study

- To improve the competency level of nurse's leaders following educational program (one year program)
- To evaluate the effectiveness of the leadership program before and after the intervention (Pre-post)
- to Evaluate the satisfaction of the participants, post the program

The sample

Before (pre-test)=48 participants

After(post-test) =46 participants

Demographic variables

From pre-test

	Categories	Frequency	Percentage
Age	30-34	17	35.4
	35-39	20	41.7
	40-44	8	16.7
	45-49	2	4.2
	50-55	1	2.1
Nationality	Non-Omani	6	12.5
	Omani	42	87.5
Gender	Female	43	89.6
	Male	5	10.4
Total years of working experience	5-9	6	12.5
	10-14	14	29.2
	15-19	19	39.6
	20-24	7	14.6
	25-30	2	4.2
Current designation	Head of Nursing/supervisor	6	12.5
	Nurse-in charge	23	47.9
	Senior staff nurse	19	39.6
Do you have post Basic specialty (Health service management)?	No	35	72.9
	Yes	13	27.1
Have you attended any type of training on leadership and management previously?	No	32	66.7
	Yes	16	33.3
My knowledge and skills in leadership & management is	Excellent	4	8.3
	Fair	7	14.6
	good	16	33.3
	Poor	2	4.2

	Very good	19	39.6
How satisfied or dissatisfied are you with Leadership & management program?	Completely satisfied	19	39.6
	Somewhat satisfied	8	16.7
	Very satisfied	19	39.6

Satisfaction scale

	Strongly Agree	Agree	Netural
Program satisfaction scale SSC)) [In my opinion the program objectives were up to my expectation]"	24	18	4
	52.2%	39.1%	8.7%
Program satisfaction scale SSC)) [The educational methods were well suited to the objectives and content]"	26	17	3
	56.5%	37.0%	6.5%
Program satisfaction scale SSC)) [The training method was used enabled us to take an active part in training]"	23	20	3
	50.0%	43.5%	6.5%
Program satisfaction scale SSC)) [The content was up to date & evidence based]"	25	19	2
	54.3%	41.3%	4.3%
Program satisfaction scale SSC)) [The training program enabled me to share professional experiences with colleagues]"	26	18	2
	56.5%	39.1%	4.3%
Program satisfaction scale SSC)) [The training program is useful for my personal development]"	29	16	1
	63.0%	34.8%	2.2%
Program satisfaction scale SSC)) [The program provided various methods of presentations and activities]"	28	17	1
	60.9%	37.0%	2.2%
Program satisfaction scale SSC)) [The training program was interactive and engaging]"	25	18	3
	54.3%	39.1%	6.5%
Program satisfaction scale SSC)) [The quality of this training session was high]"	23	19	4
	50.0%	41.3%	8.7%
Program satisfaction scale SSC)) [The program schedule was flexible]"	20	20	6
	43.5%	43.5%	13.0%
Program satisfaction scale SSC)) [The teaching materials (audio-visual) were useful]"	23	20	3
	50.0%	43.5%	6.5%
Program satisfaction scale SSC)) [Th trainer language was understood]"	28	17	1
	60.9%	37.0%	2.2%

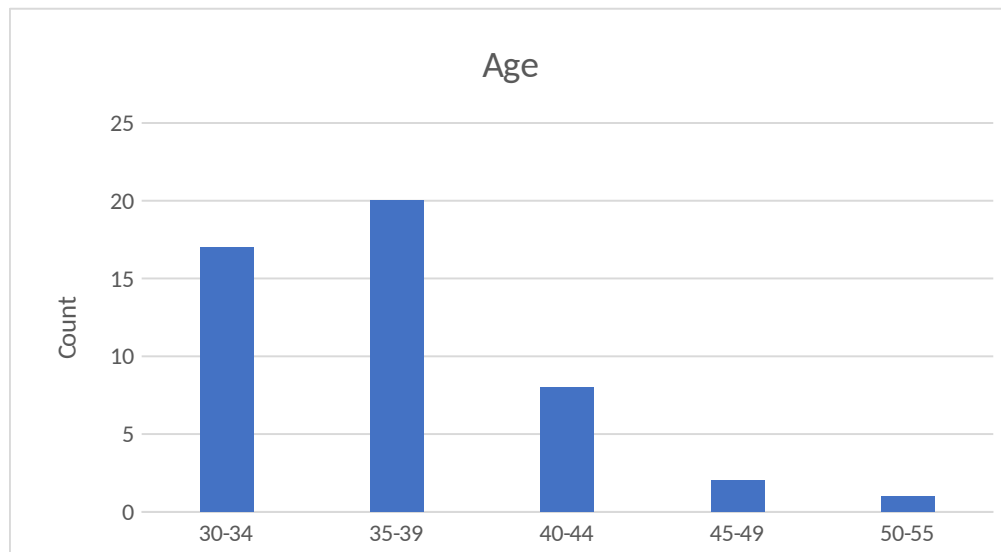
Program satisfaction scale SSC)) [The level of trainers was high which demonstrated expertise in the subject]"	30	15	1
	65.2%	32.6%	2.2%
Program satisfaction scale SSC)) [The registration process was very smooth]"	28	15	3
	60.9%	32.6%	6.5%
Program satisfaction scale SSC)) [The program was relevant to participant roles and responsibilities]"	25	19	2
	54.3%	41.3%	4.3%
Program satisfaction scale SSC)) [The program objectives and content were explained well by the organisers]"	29	15	2
	63.0%	32.6%	4.3%
Program satisfaction scale SSC)) [The leadership program improved my competency in leadership and management]"	29	13	4
	63.0%	28.3%	8.7%
Program satisfaction scale SSC)) [The program has provided good proportion of theoretical knowledge]"	25	20	1
	54.3%	43.5%	2.2%
Program satisfaction scale SSC)) [The training program enabled me to integrate the learned topic into clinical areas]"	27	16	3
	58.7%	34.8%	6.5%
Program satisfaction scale SSC)) [the program provided new information to me]"	27	17	2
	58.7%	37.0%	4.3%
Program satisfaction scale SSC)) [I am able to	25	17	4
	54.3%	37.0%	8.7%
Program satisfaction scale SSC)) [The discussion / questions during the program stimulated my critical thinking]"	21	24	1
	45.7%	52.2%	2.2%
Program satisfaction scale SSC)) [The pre / post self- assessment competency accurately assessed what I have learned in this course]"	24	19	3
	52.2%	41.3%	6.5%
Program satisfaction scale SSC)) [I have created changes after attending this training program]"	27	18	1
	58.7%	39.1%	2.2%
Program satisfaction scale SSC)) [I would recommend this course to a colleague.]"	28	17	1
	60.9%	37.0%	2.2%
Program satisfaction scale SSC)) [the program increased personal self-growth over time after training.]"	27	17	2
	58.7%	37.0%	4.3%
Program satisfaction scale SSC)) [The leadership program raised my awareness about leadership roles and responsibilities]"	30	14	2
	65.2%	30.4%	4.3%
Program satisfaction scale SSC)) [The program improved my self awareness]"	32	12	2
	69.6%	26.1%	4.3%

Data Analysis:

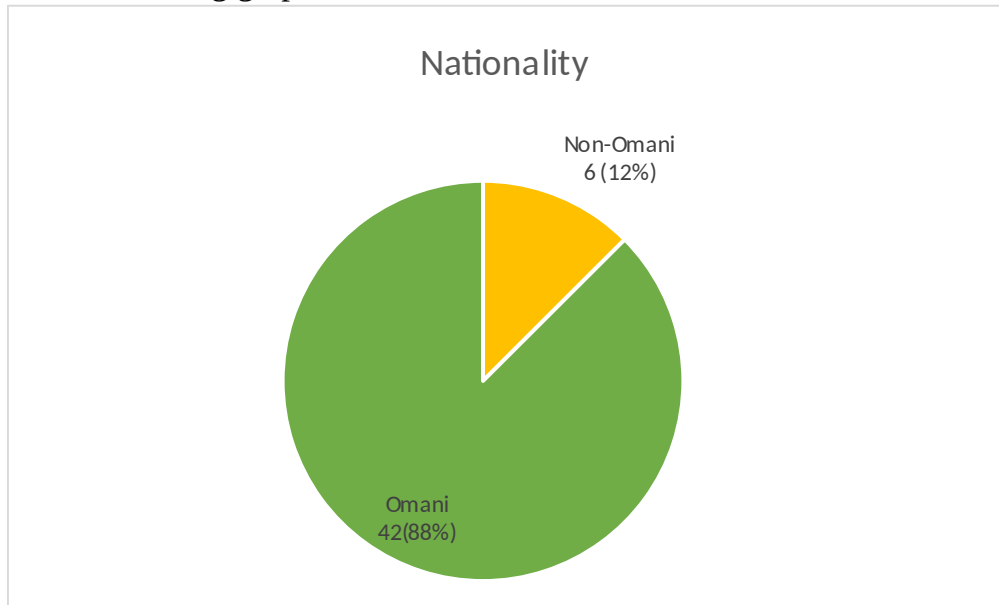
- statistically analyzed using SPSS for Windows version.
- The reliability (internal consistency) test for the questionnaires used in the study was calculated by Cronbach's alpha coefficient test.
- mean $\pm$ standard deviation (SD) for pre-post test (😊)
- The chi-square test (or Fisher's exact test when applicable) was used for comparison of variables with
- Tool reliability
- A significant level value was considered when the p-value ≤ 0.05 . Cohen's d used to measure the effect size was
- Un paired t-test as the participants are not equal in pre-test (48) and post test (46) (😊)
- ANOVA test to assess the effectiveness of the educational program (😊)

Descriptive statistics:

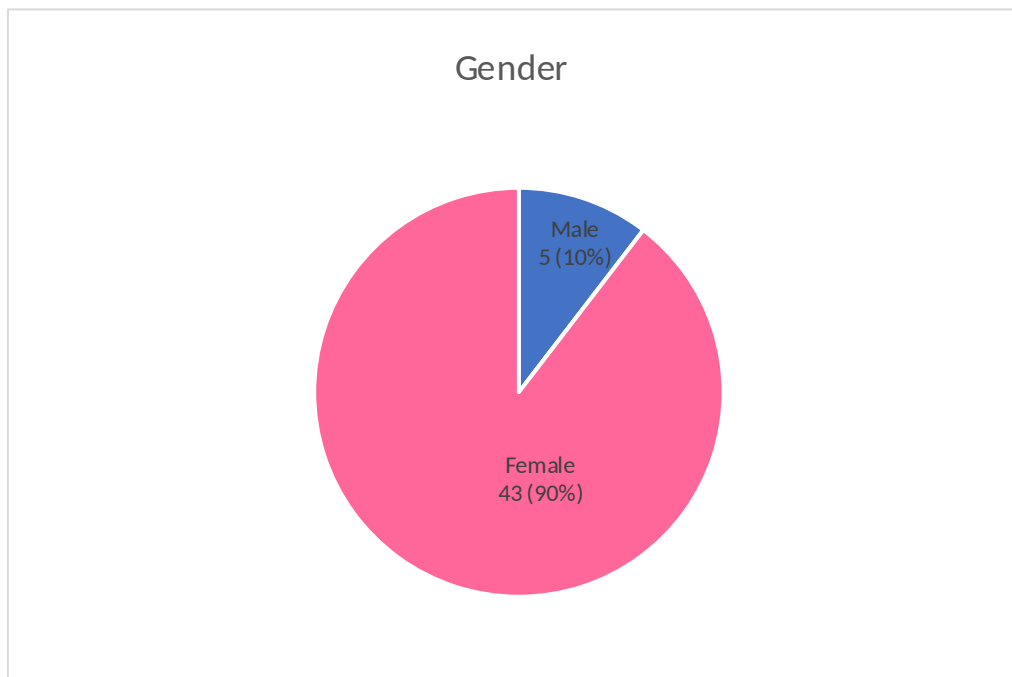
1. Age: The age ranges between 30 and 51. Its mean is 37.9, and its standard deviation is 4.57. The age distribution is left-skewed which means that it tends to lower values i.e., ages between 30 and 40 as shown in the following graph:



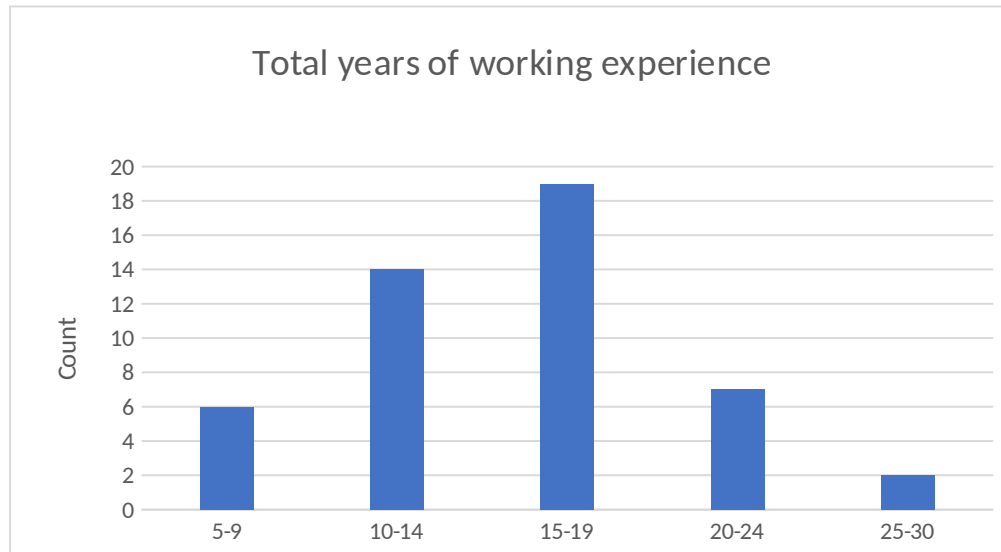
2. Nationality: Most of the participants in this study are Omanis i.e., 88%, however, there were some non-Omanis as well representing 12% of the total sample as shown in the following graph:



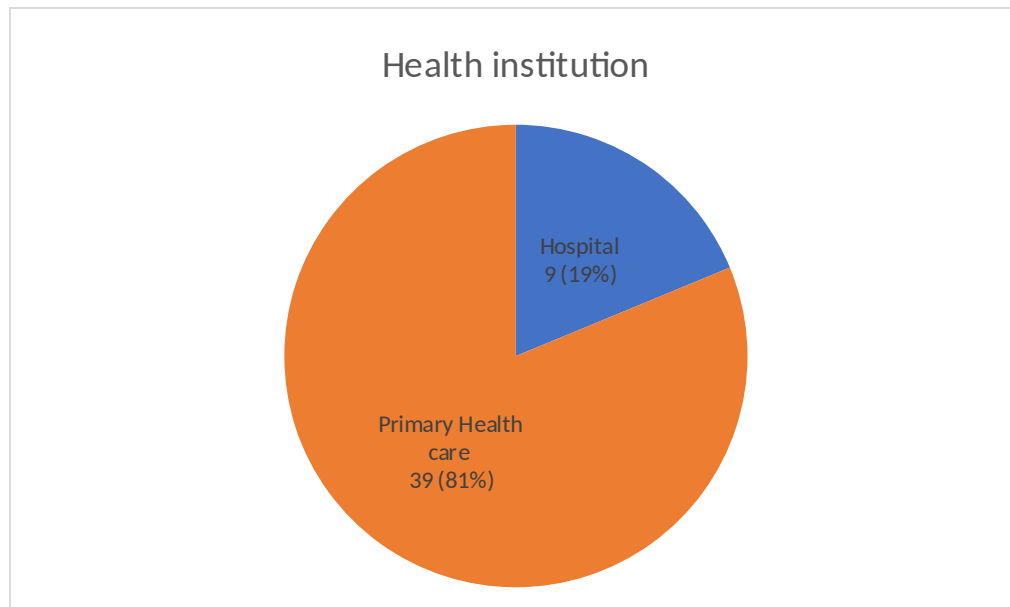
3. Gender: The demographic distribution of this research was more female-oriented as females constituted 90% of the sample, while males constituted only 10% as given in the bellow graph:



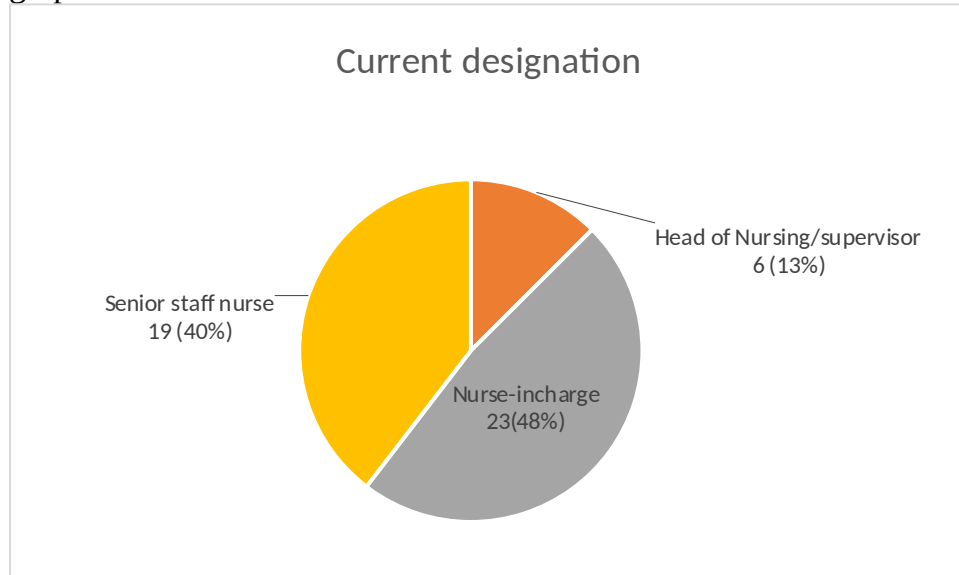
4. Total years of working experience: the total years of working experience ranges between 9 and 27 which indicate that participants of these study are experienced employees. The mean of the total years of working experience of all participants is 16.32 years while the standard deviation is 4.6. The Total years of working experience is distributed as shown in the following graph:



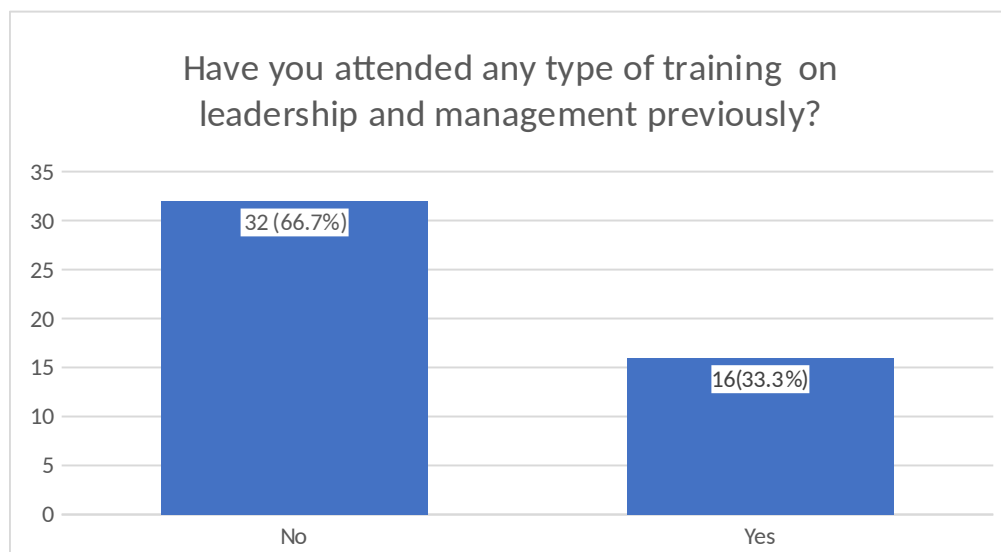
5. Health institution: most of the participants were from primary health care (81.3%) however, there were some participants from hospitals (18.8%).



6. Current designation: the most common designation among the participants are Nurse-incharge (48%), then Senior staff nurses (40%). The head of Nursing/supervisor were the least common (12%). As presented in the following graph:



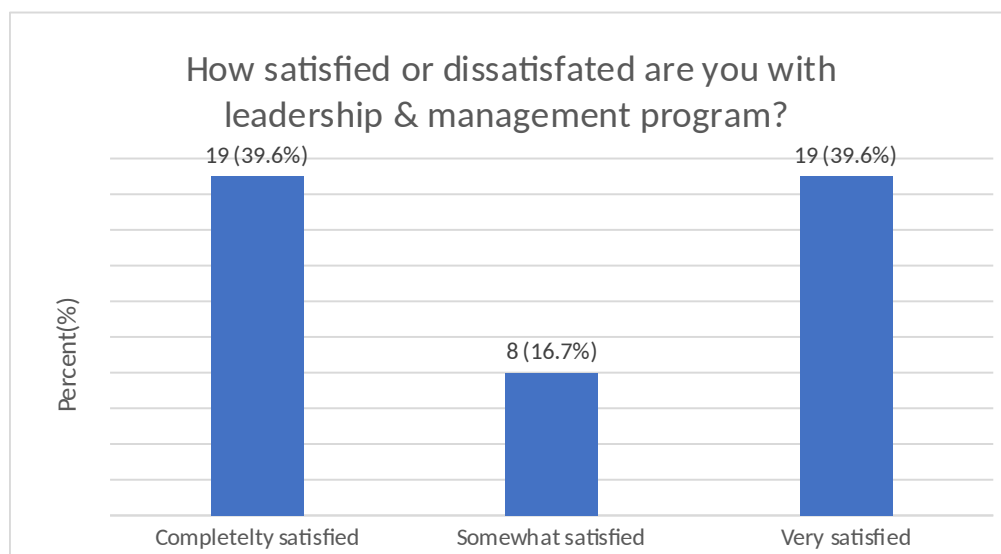
7. Have you attended any type of training on leadership and management previously?: participants who never attended any type of training on leadership and management were almost twice who attended before as shown in the following graph:



8. My knowledge and skills in leadership and management: The answers on this question from participants was heavily focusing on very good (33.3%) and good (39.6%). Percentages of other responses were also presented in the following graph:



9. How satisfied or dissatisfied are you with Leadership & Management program? Most of the participants were completely satisfied or at last very satisfied from the course (around 77%). However, there are still some participants who are either satisfied or somewhat satisfied which represent less than 20% or the total number of participants as shown in the coming graph.



Factor	Sub-factor	Pre		Post	
		Mean	Std. Deviation	Mean	Std. Deviation
Self- assessment	Developing operational objectives in SMART way and evaluate its achievement	1.77	0.592	2.00	0.667
	Strategic thinking (defining vision, mission and strategies	1.85	0.618	2.11	0.674
	Developing annual leaves plan with accurate distribution	2.29	0.617	2.61	0.537
	Succession planning- Includes developing leadership capacity of staff	1.69	0.624	2.39	0.493
	Conduct SWOT analysis effectively	1.63	0.640	2.28	0.621
	Participates in developing disaster plan for the health institutions			2.50	0.587
Human resource management	Develops proper duty Rota plan based up on MOH policy	2.46	0.651	2.63	0.488
	Distribute staff effectively, considering skill mix, patient staff ratio	2.56	0.580	2.61	0.537
	Delegates roles of the staff capabilities in distributing job duties	2.06	0.665	2.63	0.488
	Manages staff shortage	2.40	0.676	2.65	0.566
	Identifies staff educational needs , maintains continuous staff development /training activities	2.13	0.606	2.48	0.547
	Monitors staff performance effectively	1.94	0.665	2.50	0.506
	Provides solution for disagreement among staff (conflict resolution)	1.81	0.704	2.52	0.586
Service management	Demonstrates knowledge of current nursing practice, roles and functions of the team	1.88	0.703	2.15	0.556
	knowledgeable about nursing models and their application into clinical practice (e.g Nursing care plan)	1.90	0.627	2.17	0.529
	Manages clinical area during emergency/disaster	2.15	0.618	2.35	0.640
	Manages Key performance indicators	1.69	0.624	1.89	0.767
	Evaluates staff performance effectively	1.90	0.660	2.26	0.535
	Understands payment issues that affect the finance of the organizations	1.79	0.683	2.02	0.649
	Maintains patient safety measures/programs by ensuring safety compliance	2.06	0.598	2.37	0.532
	Manages risk in clinical area effectively	2.06	0.665	2.37	0.488
	Utilizes root cause analysis to solve problem/incidents	1.52	0.583	2.20	0.542
	Manages resources/equipments effectively	1.88	0.640	2.26	0.575

	Utilizes evidences for establishment of standards, practices and patient care in the organization	1.92	0.710	1.98	0.614
Personal &	Manages work/staff under stressful situation	2.27	0.536	2.54	0.504
	Seeks opportunities to learn and develop	2.19	0.607	2.54	0.585
	Able to act in an open, honest and inclusive manner - respecting other people's culture, beliefs and abilities	2.38	0.640	2.61	0.493
	Understanding current job description/ requirements	2.25	0.668	2.57	0.544
	Able to develop personal and professional careers	2.02	0.601	2.46	0.546
	Able to manage ethical dilemmas	2.04	0.713	2.43	0.583
	advocates for staff/patients rights	2.25	0.601	2.72	0.455
Leadership and managerial skills	Has coaching and guiding skills – Includes modeling behaviors of leadership and developing staff as mentors ²	1.94	0.697	2.30	0.553
	Has Decision making skills – Includes use of data driven decision making profiles/models	2.00	0.684	2.37	0.488
	Has Problem solving skills – Includes defined models for problem solving	1.96	0.713	2.41	0.541
	Has Clinical leadership thinking & skills	1.96	0.683	2.41	0.498
	Being as change agent by promoting change within the practical area)	1.79	0.713	2.41	0.580
	Has Emotional intelligence skills – Includes how well you know yourself and how you relate effectively with your environment.	2.04	0.713	2.39	0.537
	Has Self-awareness identity – Understanding one's values, beliefs, and attitudes and how they affect your responses and behaviors	2.08	0.679	2.46	0.504
	Able to work in team dynamics – Understanding the functions of group process. for both intradisciplinary/multidisciplinary groups	2.15	0.618	2.48	0.547
	Being an innovative and creative leader	2.04	0.713	2.37	0.572
Communication	Able to create a trusting environment for the staff	2.33	0.630	Not applicable	
	Able to build collaborative/ productive relationships with other teams	2.38	0.606	2.59	0.498
	Assert views in non-threatening, non-judgmental way	2.10	0.555	2.70	0.465

	Provide an environment conducive to opinion sharing, exploration of ideas and achievement of outcomes	2.21	0.549	2.48	0.505
	Provide an environment conducive to opinion sharing, exploration of ideas and achievement of outcomes	2.06	0.639	2.50	0.506

Statistical analysis:

To test if there is a significant difference between the participants answers before and after having the course, the Wilcoxon test was used as an alternative to the paired samples t.test as the assumptions of the pared samples t.test were not satisfied in the given dataset.

Factor	Sub-factor	Z	Asymp. Sig. (2-tailed) P.value	Significance of the difference
Self-assessment	Developing operational objectives in SMART way and evaluate its achievement	-1.715	0.086	Not significant
	Strategic thinking (defining vision, mission and strategies	-2.078	0.038	Significant
	Developing annual leaves plan with accurate distribution	-2.419	0.016	Significant
	Succession planning- Includes developing leadership capacity of staff	-4.398	0.000	Significant
	Conduct SWOT analysis effectively	-4.284	0.000	Significant
	Participates in developing disaster plan for the health institutions	Not applicable		
Human resource management	Develops proper duty Rota plan based up on MOH policy	-1.486	0.137	Not significant

	Distribute staff effectively, considering skill mix, patient staff ratio	-0.393	0.694	Not significant
	Delegates roles of the staff capabilities in distributing job duties	-3.846	0.000	Significant
	Manages staff shortage	-2.233	0.026	Significant
	Identifies staff educational needs , maintains continous staff development /training activities	-2.731	0.006	Significant
	Monitors staff performance effectively	-3.888	0.000	Significant
	Provides solution for disagreement among staff (conflict resolution)	-4.044	0.000	Significant
Service management	Demonstrates knowledge of current nursing practice, roles and functions of the team	-2.28	0.000	Significant
	knowledgeable about nursing models and thier application into clinical practice (e.g Nursing care plan)	-2.272	0.023	Significant
	Manages clinical area during emergency/disaster	-1.703	0.088	Not significant
	Manages Key performance indicators	-1.515	0.130	Not significant
	Evaluates staff performace effectively	-2.797	0.005	Significant
	Understands payment issues that affect the finance of the organizations	-1.847	0.065	Not significant
	Maintains patient safety measures/programs by ensuring safety compliance	-2.48	0.013	Significant
	Manages risk in clinical area effectively	-2.619	0.009	Significant
	Utilizes root cause analysis to solve problem/incidents	-4.559	0.000	Significant
	Manages resources/equipments effectively	-3.18	0.001	Significant
	Utilizes evidences for establishment of standards, practices and patient care in the	-0.468	0.640	Not significant

	organization			
Personal & amp	Manages work/staff under stressful situation	-2.711	0.007	Significant
	Seeks opportunities to learn and develop	-3.266	0.001	Significant
	Able to act in an open, honest and inclusive manner - respecting other people's culture, beliefs and abilities	-2.134	0.033	Significant
	Understanding current job description/ requirements	-2.541	0.011	Significant
	Able to develop personal and professional careers	-3.321	0.001	Significant
	Able to manage ethical dilemmas	-2.786	0.005	Significant
	advocates for staff/patients rights	-4.009	0.000	Significant
Leadership and managerial skills	Has coaching and guiding skills – Includes modeling behaviors of leadership and developing staff as mentors ²	-2.615	0.009	Significant
	Has Decision making skills – Includes use of data driven decision making profiles/models	-2.874	0.004	Significant
	Has Problem solving skills – Includes defined models for problem solving	-3.461	0.001	Significant
	Has Clinical leadership thinking & skills	-3.528	0.000	Significant
	Being as change agent by promoting change within the practical area)	-3.943	0.000	Significant
	Has Emotional intelligence skills – Includes how well you know yourself and how you relate effectively with your environment.	-3.272	0.001	Significant
	Has Self-awareness identity – Understanding one's values, beliefs, and attitudes and how they affect your responses and behaviors	-2.666	0.008	Significant

	Able to work in team dynamics – Understanding the functions of group process. for both intradisciplinary/multidisciplinary groups	-2.785	0.005	Significant
	Being an innovative and creative leader	-2.745	0.006	Significant
Communication	Able to create a trusting environment for the staff	-2.449	0.014	Significant
	Able to build collaborative/ productive relationships with other teams	-3.273	0.001	Significant
	Assert views in non-threatening, non-judgmental way	-3.157	0.002	Significant
	Provide an environment conducive to opinion sharing, exploration of ideas and achievement of outcomes	-3.266	0.001	Significant
	Provide an environment conducive to opinion sharing, exploration of ideas and achievement of outcomes	-3.439	0.001	Significant

As shown in the previous table, considering each of the considered factors the course had significant improvements on the following criteria:

1. self-assessment factor: there were significant improvements in Strategic thinking (defining vision, mission and strategies, developing annual leaves plan with accurate distribution, Succession planning- Includes developing leadership capacity of staff, Conduct SWOT analysis effectively and participates in developing disaster plan for the health institutions. However, the improvement in Developing operational objectives in SMART way and evaluate its achievement was not significant.
2. Human resource management: there were significant improvements in delegating roles of the staff capabilities in distributing job duties, managing staff shortage, Identifying staff educational needs , maintaining continuous staff development /training activities, Monitoring staff performance effectively, Providing solution for disagreement among staff (conflict resolution). On the other side, there was no significant improvements in developing proper duty Rota plan based up on MOH policy, distributing staff effectively, and considering skill mix, patient staff ratio.
3. Service management: there were significant improvements in demonstrating knowledge of current nursing practice, roles and functions of the team, knowledgeable about nursing models and their application into clinical practice (e.g

- nursing care plan), evaluating staff performance effectively, Maintaining patient safety measures/programs by ensuring safety compliance, Managing risk in clinical area effectively, Utilizing root cause analysis to solve problem/incidents, Managing resources/equipments effectively. However, the improvement in managing clinical area during emergency/disaster, managing key performance indicators, understanding payment issues that affect the finance of the organizations and utilizing evidences for establishment of standards, practices and patient care in the organization were not significant.
4. Personal & amp: there were significant improvements in managing work/staff under stressful situation seeking opportunities to learn and develop, ability to act in an open, honest and inclusive manner - respecting other people's culture, beliefs and abilities, understanding current job description/ requirements, ability to develop personal and professional careers and Ability to manage ethical dilemmas, advocates for staff/patients' rights.
 5. Leadership and managerial skills: there were significant improvements in coaching and guiding skills – including modeling behaviors of leadership and developing staff as mentors decision making skills, problem solving skills – including defined models for problem solving clinical leadership thinking, being as change agent by promoting change within the practical area, emotional intelligence skills, self-awareness identity, understanding one's values, beliefs, and attitudes and how they affect your responses and behaviors, ability to work in team dynamics, understanding the functions of group process. for both intradisciplinary/multidisciplinary groups, being an innovative and creative leader.
 6. Communication: there were significant improvements in the ability to create a trusting environment for the staff Able to build collaborative/ productive relationships with other teams, asserting views in non-threatening, non-judgmental way, providing an environment conducive to opinion sharing, exploring ideas and achievement of outcomes, providing an environment conducive to opinion sharing, exploration of ideas and achievement of outcomes

To evaluate the program satisfaction, the following questions were asked for the participants to evaluate the program using Likert scale, the agreement level was detected depending on the following rule:

Agreement level	Score	Criteria
Strongly disagree	1	1.00-1.80
Disagree	2	1.81-2.60
Nether agree or disagree	3	2.61-3.40
Agree	4	3.41-4.20
Strongly Agree	5	4.21-5.00

The average was then calculated for the participants' answers on each of these questions, as presented in the following table, after that the agreement level was detected for each question depending on the criteria presented in the previous rule:

Program satisfaction scale SSC	Average score	Agreement level
In my opinion the program objectives were up to my expectation	4.466666667	Strongly agree
The length of the course was adequate for the objectives and content	4.177777778	Agree
The structure of the program (phases method over one year) was effective to grabs the content of the program	4.2	Agree
The educational methods were well suited to the objectives and content	4.511111111	Strongly agree
The training method was used enabled us to take an active part in training	4.466666667	Strongly agree
The content was up to date & evidence based]	4.511111111	Strongly agree
The training program enabled me to share professional experiences with colleagues]	4.533333333	Strongly agree
The training program is useful for my personal development]	4.622222222	Strongly agree
The program provided various methods of presentations and activities]	4.6	Strongly agree
The training program was interactive and engaging]	4.488888889	Strongly agree
The quality of this training session was high]	4.444444444	Strongly agree
The program schedule was flexible]	4.333333333	Strongly agree
The teaching materials (audio-visual) were useful]	4.444444444	Strongly agree
Th trainer language was understood]	4.6	Strongly agree
The level of trainers was high which demonstrated expertise in the subject]	4.644444444	Strongly agree
The registration process was very smooth]	4.533333333	Strongly agree

The program was relevant to participant roles and responsibilities]	4.511111111 1	Strongly agree
The program objectives and content were explained well by the organisers]	4.6	Strongly agree
The leadership program improved my competency in leadership and management]	4.57777777 8	Strongly agree
The program has provided good proportion of theoretical knowledge]	4.53333333 3	Strongly agree
The training program enabled me to integrate the learned topic into clinical areas]	4.55555555 6	Strongly agree
the program provided new information to me]	4.55555555 6	Strongly agree
I am able to apply what I learned in clinical area]	4.48888888 9	Strongly agree
The discussion / questions during the program stimulated my critical thinking]	4.44444444 4	Strongly agree
The pre / post self- assessment competency accurately assessed what I have learned in this course]	4.46666666 7	Strongly agree
I have created changes after attending this training program]	4.57777777 8	Strongly agree
I would recommend this course to a colleague.]	4.6	Strongly agree
the program increased personal self-growth over time after training.]	4.55555555 6	Strongly agree
The leadership program raised my awareness about leadership roles and responsibilities]	4.62222222 2	Strongly agree
The program improved my self awareness]	4.66666666 7	Strongly agree
Total	4.51111111 1	Strongly agree

The average of the average scores was calculated as presented in the previous table which is 4.5. According to the rule mentioned earlier the final evaluation of program according to all the evaluation question is “strongly agree” on all of these questions.

The average score of the participants' answers were used to compare their answers on the assessment before and after having the program using the Wilcoxon Signed Ranks Test. The latter test is a nonparametric test that is alternative to the paired samples t- test which is used when the assumptions of the paired t-test are not satisfied in the dataset. In this case the normality assumption is not satisfied in the post average scores.

The Wilcoxon Signed Ranks Test showed that there is a significant difference between the average score of the participants' answers before and after having the program for all aspects as given below:

For Self-assessment planning there is a significant difference between the mean scores of pre and post having the training program ($Z=-4.6$, $p<0.05$) where the mean score of the pre and post of Self-assessment planning is ($M_{pre}=1.85$, $M_{post}=2.32$) respectively.

Human resource management also had significant difference between the pre and post responses where ($Z=-3.8$, $p<0.05$) mean score of the pre and post is ($M_{pre}=2.19$, $M_{post}=2.57$ 2.32).

For the Service management there was also had significant difference between the pre and post responses where ($Z=-3.31$, $p<0.05$) mean score of the pre and post is ($M_{pre}=1.88$, $M_{post}=2.18$).

Personal & professional Qualities also had significant difference between the pre and post responses where ($Z=-3.87$, $p<0.05$) mean score of the pre and post is ($M_{pre}=2.20$, $M_{post}=2.55$).

Leadership and managerial skills also had significant difference between the pre and post responses where ($Z=-3.65$, $p<0.05$), where the mean score of the pre and post is ($M_{pre}=2.0$, $M_{post}=2.4$).

Communication also had significant difference between the pre and post responses where ($Z=-3.72$, $p<0.05$), where the mean score of the pre and post is ($M_{pre}=2.22$, $M_{post}=2.57$).

The overall score consequently had a significant difference between the pre and post responses ($Z=-4.201$, $p<0.05$), where the mean score of the pre and post is ($M_{pre}=2.03$, $M_{post}=2.40$).